SuddWatch Phase 1: Preprocessing and Feature Engineering Strategy for County-Level Flood Forecasting in South Sudan

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***Abstract*—This report audits the raw dataset for SuddWatch, a binary classifier that forecasts whether each of South Sudan's 79 counties will exceed its own normal surface-water level 13–24 days ahead, and defines the preprocessing strategy before any production code is fitted. The panel was built in Google Earth Engine from Sentinel-1 radar, CHIRPS rainfall, SRTM, MERIT Hydro and JRC Global Surface Water, and contains N = 23,305 county-window observations and P = 26 columns (20 candidate features). Missing values (1.46% of cells) are confined to Sentinel-1-derived columns and are diagnosed as missing at random, driven by the satellite acquisition plan after the loss of Sentinel-1B. The report specifies skewness-backed imputation, distribution-matched outlier rules, a feature–target test matrix, a $|\rho| \geq 0.85$ / VIF > 10 collinearity protocol, cardinality-matched encoding, and a temporal 73.5/13.9/12.6 split whose transformers are fitted on training data only.**

***Index Terms*—flood forecasting, Sentinel-1, data preprocessing, missing data, temporal validation, South Sudan.**

I. Introduction

Recurrent flooding in the Sudd wetlands has become a leading multiplier of South Sudan's hunger and displacement crises; by October 2025 floods had affected over one million people across 29 counties [1]. Planners need to know *which county* will flood in the next few weeks, early enough to pre-position food and boats before roads and river routes close. SuddWatch frames this as supervised binary classification on a county × time panel (Table I). This Phase 1 report follows the assignment guidelines: Section III audits the dataset (Guideline A), Section IV defines cleaning, imputation, outlier and association strategy (Guideline B), and Sections V–VI define encoding, scaling, partitioning and leakage prevention (Guideline C).

Table I
Problem Specification

Item	Specification
Objective	Forecast, per county, whether surface-water extent will exceed that county's normal high-water level **13–24 days after the forecast is issued** (SDG 13 → SDG 2, SDG 1).
Task type	**Binary classification** (supervised, short-horizon spatio-temporal panel).
Unit of observation	One county × one 12-day Sentinel-1 window (12 days = Sentinel-1A repeat cycle); forecast issued at the end of window *w*.
Target Y (`y_flood_2_4wk`)	1 if the Sentinel-1 surface-water fraction in window *w*+2 exceeds the county's own **75th percentile, computed on training windows only** (floor: 1 km²). A county-specific threshold is needed because baseline wetness differs by orders of magnitude between the permanent Sudd swamps and the Equatoria highlands.
Raw panel	**N = 23,305 × P = 26** (20 candidate features, 2 IDs, 2 datetimes, 1 split tag, 1 target); issue dates 2017-01-12 → 2026-08-29.

II. Data Sources and Label Construction

A. Sources

All sources are open. Sentinel-1 GRD [2], CHIRPS daily rainfall [3], SRTM [4], MERIT Hydro [5] and JRC Global Surface Water [6] were reduced to county level in Google Earth Engine [7] using OCHA COD-AB county boundaries [8]. UNOSAT VIIRS water extents for 2022–2024 [9] are used only as an independent check on the label, never as a model input.

B. Label Construction

Sentinel-1 provides radar backscatter, not flood maps. Surface water is defined as per-pixel median VV backscatter in the 12-day window below −18 dB, restricted to terrain within 20 m of the nearest drainage (MERIT HAND), aggregated at 200 m. The −18 dB threshold was **calibrated rather than assumed**: among −14 to −20 dB it gave the best county-level agreement with UNOSAT (Spearman 0.70 on 2023-09-26).

Validation against UNOSAT (2022–2024, 1,657 county-date pairs). Across counties agreement is good (pooled Spearman ρ = **0.694**) and the national seasonal cycle agrees moderately (ρ = 0.379 over 27 dates). *Within* counties over time, the comparison

that matters for a county-relative target, agreement is uneven: median ρ by state Jonglei 0.51, Central Equatoria 0.39, Lakes 0.18, Unity 0.17, Eastern Equatoria -0.01, Upper Nile -0.03, Western Equatoria -0.14, Warrap -0.46, Western Bahr el Ghazal -0.47; overall median 0.041. It is strongest in Jonglei, the core Sudd state, and weak or negative in drier, more vegetated western and northern counties. Causes: (i) SAR detects open water but misses water under dense papyrus and reed canopies, whereas VIIRS (375 m) also captures partly flooded vegetation; (ii) 2022–2024 is the single-satellite Sentinel-1 period with the sparsest coverage. **Phase 2 actions:** add VH-based flooded-vegetation detection, run a −16 dB sensitivity variant, and report metrics nationally and for the Sudd core (Jonglei, Unity, Upper Nile, Lakes).

III. Dataset and Problem Audit (Guideline A)

A. Initial Shape

The raw, un-preprocessed panel `data/raw/suddwatch_panel_unprocessed.csv` has **N = 23,305 observations** (79 counties × 295 issue windows) and **P = 26 columns**, of which **20 are candidate predictive features**.

B. Variable Inventory

Counts by type: Continuous: 17, Datetime: 2, Nominal (ID / join key): 1, Nominal (ID / label): 1, Nominal categorical: 1, Nominal (partition tag, not a feature): 1, Discrete (count): 1, Discrete (cyclical): 1, Binary target (Y): 1. **No ordinal variables exist**: states have no rank, and month is cyclical (December is adjacent to January), so an ordinal code would impose a false 12-to-1 distance. Table II lists every column.

Table II
Variable Inventory and Baseline Missingness

column	type	role	dtype	n_unique	n_missing	pct_missing
adm2_pcode	Nominal (ID / join key)	identifier/meta	str	79	0	0.00
adm2_name	Nominal (ID / label)	identifier/meta	str	79	0	0.00
adm1_name	Nominal categorical	feature	str	11	0	0.00
window_start	Datetime	identifier/meta	datetime64[us]	295	0	0.00
issue_date	Datetime	identifier/meta	datetime64[us]	295	0	0.00
split	Nominal (partition tag, not a feature)	identifier/meta	str	3	0	0.00
n_scenes	Discrete (count)	feature	int64	53	0	0.00
obs_frac	Continuous	feature	float64	2820	0	0.00
water_frac	Continuous	feature	float64	16798	2145	9.20
water_frac_lag1	Continuous	feature	float64	16734	2224	9.54
water_frac_lag2	Continuous	feature	float64	16671	2303	9.88
rain_12d_mm	Continuous	feature	float64	20445	0	0.00
rain_36d_mm	Continuous	feature	float64	21652	0	0.00
rain_90d_mm	Continuous	feature	float64	22962	0	0.00
rain_90d_anom_mm	Continuous	feature	float64	23304	0	0.00
county_area_km2	Continuous	feature	float64	79	0	0.00
elev_mean_m	Continuous	feature	float64	79	0	0.00
elev_std_m	Continuous	feature	float64	79	0	0.00
slope_mean_deg	Continuous	feature	float64	79	0	0.00
hand_mean_m	Continuous	feature	float64	79	0	0.00
hand_lt5m_frac	Continuous	feature	float64	71	0	0.00
twi_mean	Continuous	feature	float64	79	0	0.00
perm_water_frac	Continuous	feature	float64	41	0	0.00
jrc_water_occurrence_pct	Continuous	feature	float64	79	0	0.00
month	Discrete (cyclical)	feature	int64	12	0	0.00
y_flood_2_4wk	Binary target (Y)	target	float64	2	2145	9.20

C. Baseline Missingness

Before any processing, **1.46%** of all 605,930 cells are missing. **19,708** rows are complete and **3,597** rows (15.4%) contain at least one missing value. Missingness is confined to the Sentinel-1-derived columns (Table III, Fig. 1).

Table III
Columns With Missing Values

column	type	n_missing	pct_missing
water_frac	Continuous	2145	9.20
water_frac_lag1	Continuous	2224	9.54
water_frac_lag2	Continuous	2303	9.88
y_flood_2_4wk	Binary target (Y)	2145	9.20

Fig 1.1 - Missingness pattern (red dashes: Sentinel-1B loss, Sentinel-1C/D coverage)

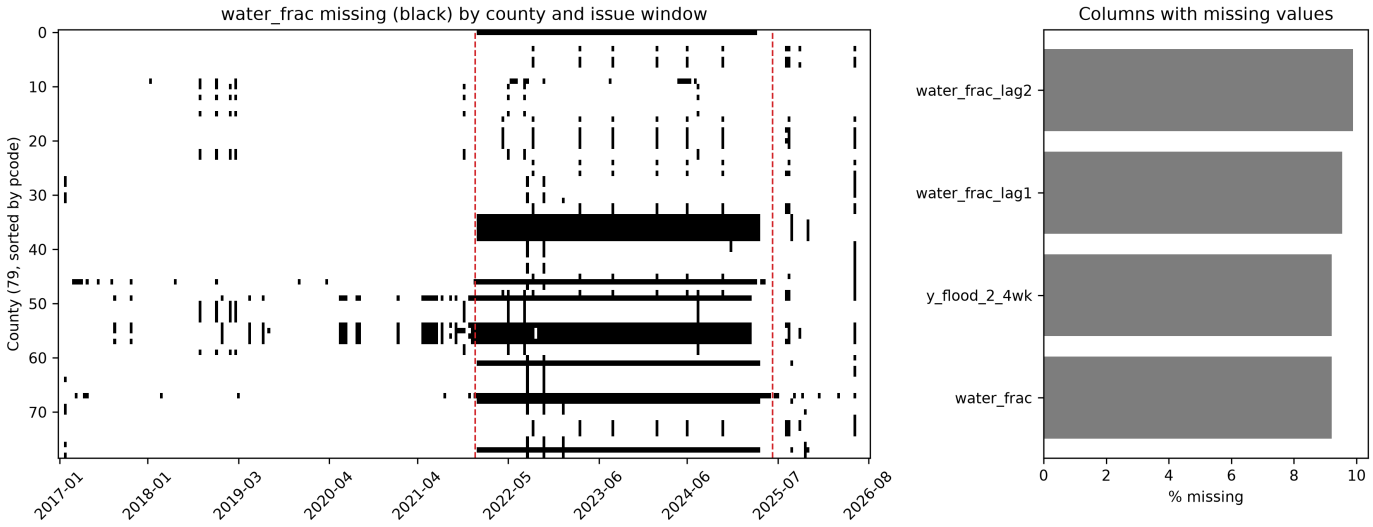


Fig. 1. Missingness pattern (assignment Fig. 1.1): `water_frac` missing by county × issue window (left) and per-column missing percentage (right). Red dashes mark the Sentinel-1B loss and the Sentinel-1C/D coverage.

D. Target Label Specification

The task is binary classification. Table IV and Fig. 2 give the class balance of labelled rows per temporal split.

TABLE IV
CLASS BALANCE BY TEMPORAL SPLIT

split	issue dates	rows	labelled	Y=1	Y=0	positive rate	neg:pos
train	2017-01-12 → 2023-12-31	16906	15550	2987	12563	19.2%	4.21:1
validation	2024-01-12 → 2025-07-05	3634	2948	1072	1876	36.4%	1.75:1
test	2025-07-17 → 2026-08-29	2765	2662	980	1682	36.8%	1.72:1

Fig 1.2 - Target class balance by temporal split

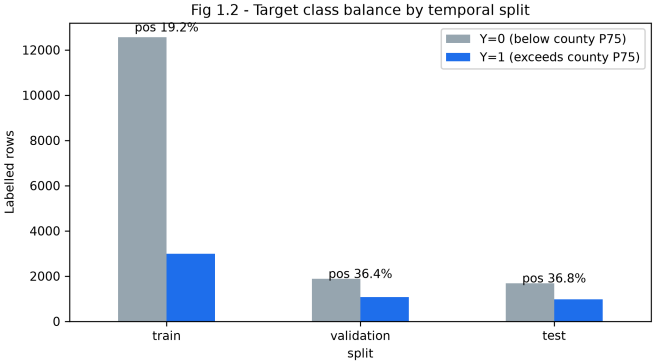


Fig. 2. Target class balance by split (assignment Fig. 1.2).

Imbalance. The training set is moderately imbalanced (4.21:1, 19.2% positive). The rate is below the nominal 25% because of the 1 km² floor: in 24 of 79 dry counties the training P75 is below 1 km² of water, so the floor binds and those counties are positive in only 6.6% of training windows, versus 25.2% in the other 55 counties (≈25% by construction). This is intended: in very dry counties, crossing a sub-1 km² level would reflect radar noise rather than flooding. Positive rates are higher in validation and test because the threshold is fixed from 2017–2023 and later seasons were wetter; the 2025 season affected over one million people [1]. A sensor effect cannot be fully excluded for the test period (Sentinel-1C/D add scenes from mid-2025); Phase 2 will check it by stratifying exceedance rates by `n_scenes`.

Mitigation. (1) `class_weight="balanced"` in the estimator (no synthetic rows, no information loss); (2) the decision threshold is chosen on the *validation* precision–recall curve to meet the proposal's <15% false-alarm target; (3) evaluation uses PR-AUC, F1 and recall rather than accuracy. If SMOTE is tried, it runs inside training folds only, after the temporal split. Random under-sampling is rejected because it discards whole flood seasons from a temporally scarce record.

IV. CLEANING AND IMPUTATION STRATEGY (GUIDELINE B)

A. Missingness Mechanism Classification

Table V assigns each column with missing values to a mechanism, with the evidence from this dataset.

TABLE V
MISSINGNESS MECHANISMS, EVIDENCE AND TREATMENT

Column(s)	Mechanism	Evidence from this dataset	Treatment
`water_frac`	**MAR**	Missing when <50% of the county was imaged. Missing share by satellite era: S1A+S1B (2017-2021): **1.6%**; S1A only (2022-mid 2025): **21.7%**; S1A+S1C/D (mid 2025-2026): **3.7%**; χ^2 test of era × missingness p < 1e-16; by state p < 1e-16 (highest: Northern Bahr el Ghazal 35.5%, Abyei Region 34.6%, Western Bahr el Ghazal 25.8%). The gap is explained by *observed* variables: date (Sentinel-1B failure on 23 Dec 2021; Sentinel-1C/D arrival in 2025) and location (orbit tracks excluded from the single-satellite plan). **Not MCAR**: strongly systematic. **Not MNAR**: acquisitions are planned in advance, independently of ground conditions. In 2022–2024 raw 36-day rainfall before missing windows is lower (median 77.2 vs 95.3 mm, Mann-Whitney p = 0.005) only because un-imaged counties are drier places: in a logistic regression of missingness on state + month + standardised rainfall, the rainfall coefficient is -0.016 (p = 0.86). MNAR can never be fully excluded from observed data alone; here it is ruled out by the acquisition mechanism and this adjusted test.	County-level **training-median** imputation + binary `_was_missing` indicator.
`water_frac_lag1`, `water_frac_lag2`	**MAR** (+ structural)	Shifted copies of `water_frac`, so they inherit its gaps; the first 1–2 windows of 2017 are missing by construction.	Same imputer + indicator.
`y_flood_2_4wk`	**MAR**	Undefined when window *w*+2 was not imaged (same mechanism).	**Never imputed**; rows without Y are excluded from fitting and evaluation.
Rainfall, terrain, calendar, IDs	—	0 missing (complete gridded products; static county attributes).	None needed.

B. Imputation Choice

Median, not mean: `water_frac` is extremely right-skewed (training skewness = **22.6**; mean = 0.0037 vs median = 0.0004), so a mean fill would inject a value inflated by rare floods into ordinary dry windows. **Per county:** baseline wetness differs by orders of magnitude between counties, and a global median would erase that. **Not KNN:** gaps are spatially clustered (whole orbit tracks missing together), so a county's nearest neighbours are usually missing in the same window. **Not last-observation-carried-forward:** nine counties were not imaged for up to about three years (2022–2025), so carried values would be stale. **Mode** is inapplicable to a continuous feature. Indicator columns preserve the fact that a value was imputed.

C. Outlier Identification Rules

The rule follows distribution shape: |skew| ≤ 0.5 → Z-score (|z| > 3); |skew| > 0.5 → 1.5×IQR. Bounds in Table VI use training rows only and are re-learned inside the pipeline (`OutlierCapper.fit`) on `X_train`.

TABLE VI
SKEWNESS, OUTLIER RULES AND TREATMENT (TRAINING ROWS)

feature	skew	mean	median	rule	IQR low	IQR high	% IQR out	% |z|>3	treatment
obs_frac	-2.90	0.912	1	IQR (skewed)	0.999	1	18.7%	5.8%	Not capped: coverage metadata bounded in [0,1]; low values are real acquisition gaps.
water_frac	22.60	0.0037	0.0004	IQR (skewed)	-0.0052	0.0087	12.5%	0.9%	Flag only (IQR); NOT capped: outliers are flood events (the signal). log1p(100·x) compresses the tail.
water_frac_lag1	22.93	0.0037	0.0004	IQR (skewed)	-0.0052	0.0087	12.5%	0.9%	Same as water_frac.
water_frac_lag2	23.29	0.0037	0.0004	IQR (skewed)	-0.0052	0.0087	12.5%	0.9%	Same as water_frac.
rain_12d_mm	0.50	32.4	29.1	IQR (moderate skew)	-83.2	141	0.0%	0.2%	Cap at training 1.5×IQR fences (skewed).
rain_36d_mm	0.24	97.4	94.4	Z-score (|z|>3)	-234	412	0.0%	0.0%	Cap at training μ ± 3σ (near-normal).
rain_90d_mm	0.16	242	237	Z-score (|z|>3)	-466	927	0.0%	0.0%	Cap at training μ ± 3σ (near-normal).
rain_90d_anom_mm	0.48	7.4	1.53	Z-score (|z|>3)	-70	84.1	6.9%	1.2%	Cap at training μ ± 3σ (near-normal).
county_area_km2	4.29	8.14e+03	5.87e+03	IQR (skewed)	-3.42e+03	1.74e+04	6.3%	2.5%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
elev_mean_m	1.91	513	432	IQR (skewed)	134	856	6.3%	2.5%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
elev_std_m	3.43	50.8	16.7	IQR (skewed)	-83	147	7.6%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
slope_mean_deg	3.07	0.76	0.246	IQR (skewed)	-1.21	2.46	6.3%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
hand_mean_m	3.27	6.52	1.49	IQR (skewed)	-13.7	24.1	3.8%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
hand_lt5m_frac	-0.70	0.741	0.935	IQR (moderate skew)	-0.586	1.95	0.0%	0.0%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
twi_mean	-0.68	12.2	12.9	IQR (moderate skew)	6.73	17.3	0.0%	0.0%	Static: real geography, not errors → not capped; log1p if |skew| > 1.

feature	skew	mean	median	rule	IQR low	IQR high	% IQR out	% |z|>3	treatment
perm_water_frac	2.99	0.0009	0	IQR (skewed)	-0.0013	0.0021	15.2%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
jrc_water_occurrence_pct	2.34	0.157	0.0194	IQR (skewed)	-0.313	0.524	8.9%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.

D. Outlier Treatment: Capping, Not Deletion

No row is deleted for an outlier. (a) **Preserves N:** deleting every row with a flagged value would remove a large share of the panel and preferentially remove flood seasons, the minority class. (b) **Prevents information loss:** a row with one extreme rainfall value still carries valid terrain, calendar and water-history signal. (c) **Domain rule:** water-fraction "outliers" are the floods SuddWatch exists to predict, so they are flagged but **not capped**; static terrain extremes are real counties (e.g. the Equatoria highlands), so they are log-transformed, not capped. Only dynamic rainfall features are winsorised, at the training bounds.

E. Feature–Target Association Matrix

Continuous or discrete feature × binary Y → **point-biserial r** plus **Mann-Whitney U** (rank-based, robust to the skew in Table VI) plus **mutual information** (captures non-monotonic effects such as seasonality); nominal state × Y → **χ^2 with Cramér's V** and MI. Spearman ρ is used between features (Section IV-F) because most are non-normal. Static terrain features take only 79 distinct values repeated over time, so their row-level p-values overstate the evidence; MI is therefore the ranking and tie-break criterion (Table VII).

TABLE VII
FEATURE–TARGET ASSOCIATION (TRAINING ROWS)

feature	type	test	|r_pb| / Cramér's V	p (r_pb / χ^2)	p (Mann-Whitney)	mutual_info
water_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.257	<1e-16	<1e-16	0.1188
water_frac_lag1	Continuous	Point-biserial r + Mann-Whitney U + MI	0.239	<1e-16	<1e-16	0.0979
water_frac_lag2	Continuous	Point-biserial r + Mann-Whitney U + MI	0.216	<1e-16	<1e-16	0.0908
rain_36d_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.346	<1e-16	<1e-16	0.0779
rain_90d_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.340	<1e-16	<1e-16	0.0680
month	Discrete (cyclical)	Point-biserial r + Mann-Whitney U + MI	0.218	<1e-16	<1e-16	0.0623
rain_12d_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.311	<1e-16	<1e-16	0.0596
hand_lt5m_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.161	<1e-16	<1e-16	0.0382
county_area_km2	Continuous	Point-biserial r + Mann-Whitney U + MI	0.047	<1e-16	<1e-16	0.0381
jrc_water_occurrence_pct	Continuous	Point-biserial r + Mann-Whitney U + MI	0.091	<1e-16	<1e-16	0.0380
slope_mean_deg	Continuous	Point-biserial r + Mann-Whitney U + MI	0.121	<1e-16	<1e-16	0.0379
elev_mean_m	Continuous	Point-biserial r + Mann-Whitney U + MI	0.158	<1e-16	<1e-16	0.0379
elev_std_m	Continuous	Point-biserial r + Mann-Whitney U + MI	0.091	<1e-16	<1e-16	0.0379
hand_mean_m	Continuous	Point-biserial r + Mann-Whitney U + MI	0.121	<1e-16	<1e-16	0.0379
twi_mean	Continuous	Point-biserial r + Mann-Whitney U + MI	0.160	<1e-16	<1e-16	0.0378
rain_90d_anom_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.088	<1e-16	<1e-16	0.0354
obs_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.009	2.4e-01	1.3e-03	0.0327
n_scenes	Discrete (count)	Point-biserial r + Mann-Whitney U + MI	0.112	<1e-16	<1e-16	0.0257
adm1_name	Nominal categorical	Chi-square + Cramer's V + MI	0.169	<1e-16		0.0162
perm_water_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.071	<1e-16	<1e-16	0.0131

F. Multicollinearity Protocol

(1) Pairwise screen with |Spearman ρ| ≥ 0.85 on training rows; (2) walk pairs from the strongest |ρ| down and, if both are still present, **drop the feature with lower mutual information with Y**; (3) iterate VIF on the survivors, dropping the highest VIF > 10 until all VIF ≤ 10. Tables VIII–X give the pairs, the resulting drop list (applied via `DROP_COLLINEAR` in `src/pipeline_builder.py`) and VIF before and after.

TABLE VIII
COLLINEAR PAIRS (|ρ| ≥ 0.85)

feature_a	feature_b	spearman	dropped_by_rule
water_frac	water_frac_lag1	0.944	water_frac_lag1
water_frac	water_frac_lag2	0.926	water_frac_lag2
water_frac_lag1	water_frac_lag2	0.944	water_frac_lag1
rain_12d_mm	rain_36d_mm	0.916	rain_12d_mm
rain_36d_mm	rain_90d_mm	0.866	rain_90d_mm
elev_mean_m	elev_std_m	0.920	elev_std_m
elev_mean_m	slope_mean_deg	0.882	slope_mean_deg
elev_mean_m	hand_mean_m	0.888	hand_mean_m
elev_mean_m	hand_lt5m_frac	-0.883	elev_mean_m
elev_mean_m	twi_mean	-0.882	twi_mean
elev_std_m	slope_mean_deg	0.894	elev_std_m
elev_std_m	hand_mean_m	0.908	hand_mean_m
elev_std_m	hand_lt5m_frac	-0.851	elev_std_m
elev_std_m	twi_mean	-0.863	twi_mean
slope_mean_deg	hand_mean_m	0.960	hand_mean_m
slope_mean_deg	hand_lt5m_frac	-0.907	slope_mean_deg
slope_mean_deg	twi_mean	-0.950	twi_mean
hand_mean_m	hand_lt5m_frac	-0.950	hand_mean_m
hand_mean_m	twi_mean	-0.978	twi_mean
hand_lt5m_frac	twi_mean	0.964	twi_mean
perm_water_frac	jrc_water_occurrence_pct	0.894	perm_water_frac

TABLE IX
COLLINEARITY DROP LIST

feature	reason
twi_mean	|rho|=0.978 with hand_mean_m
hand_mean_m	|rho|=0.960 with slope_mean_deg
water_frac_lag1	|rho|=0.944 with water_frac
water_frac_lag2	|rho|=0.926 with water_frac
elev_std_m	|rho|=0.920 with elev_mean_m
rain_12d_mm	|rho|=0.916 with rain_36d_mm
slope_mean_deg	|rho|=0.907 with hand_lt5m_frac
perm_water_frac	|rho|=0.894 with jrc_water_occurrence_pct
elev_mean_m	|rho|=0.883 with hand_lt5m_frac
rain_90d_mm	|rho|=0.866 with rain_36d_mm

TABLE X
VARIANCE INFLATION FACTORS

feature	VIF_before	VIF_after
slope_mean_deg	270.75	
hand_mean_m	225.65	
twi_mean	202.42	
hand_lt5m_frac	174.91	1.28
elev_std_m	19.51	
rain_36d_mm	12.52	1.10
elev_mean_m	11.60	
jrc_water_occurrence_pct	10.66	1.26
perm_water_frac	9.63	
rain_12d_mm	6.06	
rain_90d_mm	4.92	
water_frac_lag1	1.81	
water_frac	1.77	1.10
water_frac_lag2	1.75	
county_area_km2	1.42	1.04
rain_90d_anom_mm	1.16	1.11
n_scenes	1.15	1.14
obs_frac	1.11	1.10

Retained numeric features: `obs_frac`, `water_frac`, `rain_36d_mm`, `rain_90d_anom_mm`, `county_area_km2`, `hand_lt5m_frac`, `jrc_water_occurrence_pct`, `n_scenes`. The terrain block (elevation, slope, HAND, TWI) is one physical gradient from highland to floodplain, so one representative suffices. The autoregressive signal is kept through `water_frac`, the latest observed water state two windows before the target; Phase 2 will test a less collinear momentum feature (`water_frac - water_frac_lag1`).

V. ENCODING, SCALING AND TRANSFORMATION (GUIDELINE C1)

A. Encoding Matched to Cardinality

TABLE XI
ENCODING DECISIONS

Variable	Type / cardinality	Encoding	Justification
`adm1_name` (state)	Nominal, 11 levels	One-hot (`handle_unknown="ignore"`)	Low cardinality, no order; 11 binary columns keep states distinct.
`adm2_pcode` (county)	Nominal, 79 levels	**Not encoded**: join key, imputation group and CV group only	79 one-hot columns on ~295 rows per county would let a model memorise county identity; county behaviour enters through terrain and water history.
`month`	Discrete, cyclical 1–12	sin/cos pair	Preserves Dec–Jan adjacency; an ordinal code would place them 11 units apart.

Variable	Type / cardinality	Encoding	Justification
Ordinal variables	None present	OrdinalEncoder not applicable	No variable has an inherent rank; forcing one would invent order.

B. Scaling Matched to Distributions and Algorithms

TABLE XII
TRANSFORMATION AND SCALING PLAN

Feature group	Distribution	Transform → scaler	Reason
Water fractions	Extreme right skew (≈23)	`log1p(100·x)` → StandardScaler	Log compresses the flood tail; MinMax alone would squash 99% of values near 0.
County area, elevation mean/std, slope, HAND, water occurrence	|skew| > 1	`log1p` → StandardScaler	Multiplicative scales; log brings them closer to symmetric before standardisation.
Rainfall totals and anomaly, TWI, scene count	|skew| ≤ 1	StandardScaler (after capping)	Near-symmetric; mean 0 / sd 1 required by the logistic-regression baseline (L2 penalty) and by PCA (Fig. 5.1).
`obs_frac`, `hand_lt5m_frac`	Bounded [0, 1]	MinMaxScaler	Known physical bounds; preserves exact 0 and 1.

The primary model (gradient-boosted trees) is invariant to monotone scaling, so scaling is not needed for it; it is applied for the logistic-regression baseline, PCA and any distance-based comparison. `build_preprocessor(scale_tree_inputs=False)` skips it for trees.

VI. Data Partitioning and Leakage Prevention (Guideline C2)

A. Partitioning Ratios

Temporal blocks of whole flood seasons: train = issue dates before 2024-01-01, validation = 2024-01-01 to 2025-06-30, test = 2025-07-01 to 2026-08-31. Of 21,160 labelled rows this gives **73.5% / 13.9% / 12.6%**. Seven training seasons (2017–2023) cover wet and dry years, and the held-out test year contains the full 2025 flood season named in the proposal. With N ≈ 23k an 80/20 random split would be statistically ample, but randomness is the wrong criterion for a forecasting task (Section VI-D).

B. Stratification

`StratifiedShuffleSplit` is **rejected**: it mixes future windows into training (look-ahead leakage) and, because neighbouring windows are autocorrelated, inflates scores. Class proportions are instead reported per split (Table IV) and per CV fold, and imbalance is handled by class weights and validation-tuned thresholds.

C. Grouping

Every county appears in every split by design: the deployed system forecasts the same 79 counties forward in time, so the operational question is temporal generalisation. A supplementary `GroupKFold` by county within the training period will measure generalisation to unseen counties as a robustness check.

D. Temporal Ordering and Validation Scheme

All splits move forward in time. Hyper-parameters are tuned inside training with `TimeSeriesSplit(n_splits=5, gap=2 windows × 79 rows)`: the gap purges rows whose target window (w+2) would overlap the next fold. The last two training windows are likewise purged before validation, and every preprocessing step is re-fitted inside each fold.

E. Leakage-Prevention Architecture

TABLE XIII
LEAKAGE PATHS AND CONTROLS

Leakage path	Control
Target threshold sees the future	County P75 computed on training windows only (`county_thresholds_train_only.csv`).
Rainfall after the issue date	Sums end on the issue date; anomaly climatology from 2001–2016, before the study period.
Future satellite data	Only windows w, w−1, w−2 enter X; w+1 and the target window w+2 never do.
Imputation, capping, scaling, encoding	One scikit-learn [10] `Pipeline`: `GroupMedianImputer → OutlierCapper(Z) → OutlierCapper(IQR) → ColumnTransformer → estimator`; `fit` on `X_train` only and re-fitted in every CV fold; validation/test are only transformed.
Feature selection	Collinearity and MI statistics computed on training rows only.
Same-time proxies of Y	Excluded: no same-window flood or population layers (e.g. UNOSAT "Pop exposed"); UNOSAT is used only for label validation.

VII. Conclusion

The audit shows a 23,305 × 26 panel whose only missing values follow a documented, testable MAR mechanism, a moderately imbalanced target with a stated mitigation plan, and a heavily collinear terrain block that reduces to a few representatives. Phase 2 will implement this strategy in the pipeline defined in `src/pipeline_builder.py`, produce the eight diagnostic figures, and address the label-validation limitation in Section II. All code, audit tables and figures are in the repository (https://github.com/Super-Talen/SuddWatch); every number in this report is generated by `src/audit.py` and `src/report_phase1.py`.

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